

ON LOCAL TRANSFER FACTORS FOR ENDOSCOPY THEORY

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1. NOTATIONS

1.1. Basic notations.

- F global or local field, we only consider the local case.
- $\Gamma_F = \text{Gal}(\bar{F}/F)$.
- W_F Weil group of F when local.
- G connected reductive group over F .
- $\psi : G \rightarrow G^*$ quasi-split inner form of G , $\psi \sigma(\psi)^{-1} = \text{Int}(u_\sigma)$ for some u_σ .
- Spl: splitting data for the reductive groups.
- $\hat{G}$: dual group of G , defined over $\mathbb{C}$.
- ${}^L G$: L-group of G , realized as ${}^L G = \hat{G} \rtimes W_F$, the action is specified by a splitting on $\hat{G}$.
- Borel and torus: we will denote B_H, T_H in H , B_G, T_G in G , B, T in G^* , $\hat{B}_H, \hat{T}_H$ in $\hat{H}$, $\mathcal{B}, \mathcal{T}$ in $\hat{G}$.
- Usually $\text{Spl}_{\hat{H}} = (\mathcal{B}_H, \mathcal{T}_H, \{X_\alpha\}_H)$, $\text{Spl}_{\hat{G}} = (\mathcal{B}, \mathcal{T}, \{X_\alpha\})$.

1.2. Endoscopy data. Let $(H, \mathcal{H}, s, \xi)$ be endoscopic data for G , which means:**Structure:**

- (1) H is a quasi-split group over F .
- (2) $\mathcal{H}$ is a split extension of W_F by $\hat{H}$ and $\rho_{\mathcal{H}}$ coincides with ρ_H .
- (3) s is a semisimple element of $\hat{G}$.
- (4) $\xi : \mathcal{H} \rightarrow {}^L G$ is an L -homomorphism.

Properties:

- (1) $\text{Ints} \circ \xi = a \otimes \xi$, where a is a 1-cocycle of W_F in the center $Z(\hat{G})$.
- (2) $\xi|_{\hat{H}}$ is an isomorphism of $\hat{H}$ onto $(\hat{G}^s)^0$.

For further definition, see Renard.

1.3. Admissible embeddings. The isomorphism $\hat{T}_G \rightarrow \mathcal{T}_G$ is from choice of (B_G, T_G) , $(\mathcal{B}_G, \mathcal{T}_G)$, similarly we have $\hat{T}_H \rightarrow \mathcal{T}_H$. Thus we have

$$\hat{T}_H \rightarrow \mathcal{T}_H \xrightarrow{\xi} \mathcal{T}_G \rightarrow \hat{T}_G$$

Thus an isomorphism $T_H \rightarrow T_G$. We have

$$\mathcal{A}_{H/G} : Cl_{ss}(H(\bar{F})) \rightarrow Cl_{ss}(G(\bar{F})).$$

This is canonical, since we have quotient the Weyl group. And $\mathcal{A}_{H/G}$ is a Γ_F -map (orbits).

Suppose that T_H is defined over F , if (B_{G^*}, T_{G^*}) is chosen such that T_{G^*} and $T_H \rightarrow T_{G^*}$ are defined over F , then we call $T_H \rightarrow T_{G^*}$ an admissible embedding of T_H in T_{G^*} . This is uniquely determined up to $\mathfrak{U}(T_{G^*})$ -conjugacy. (Langlands, it is usually $\text{Int}(g^{-1})$)

We call strongly G -regular $\gamma_H \in H(F)$ an *image* of $\gamma_G \in G(F)$ if γ_G lies in the image under $\mathcal{A}_{H/G}$ of the conjugacy class of γ_H in $H(\bar{F})$, the twisted analogue is called *norm*.

2. PRELIMINARIES

This and the next section are just the content of Langlands-Shelstad, with some complements and remarks.

2.1. General settings. Suppose k is a field of characteristic zero with algebraic closure $\bar{k}$, G is a connected reductive algebraic group defined and quasi-split over k .

- $(\mathbf{B}, \mathbf{T}, \{X_\alpha\})$ is a k -splitting of G .
- Γ is an abstract group, acting on G through $\text{Aut}(G, \mathbf{B}, \mathbf{T}, \{X_\alpha\})$.
- $\Omega = \Omega(G, \mathbf{T})$, Weyl group of $\mathbf{T}$ in G . It has a Γ action on it.

For $\theta = w \rtimes \rho \in \Omega \rtimes \Gamma$, define $n(\theta) = n(w) \rtimes \rho \in G(\bar{k}) \rtimes \Gamma$, where $n(w) \in N(\mathbf{T}(\bar{k}), G(\bar{k}))$ is given by the construction in Springer. We explain here.

- (1) Let α be a simple root of $\mathbf{T}$ in $\mathbf{B}$, by definition $\rho X_\alpha = X_{\rho\alpha}$, $\forall \rho \in \Gamma$. Let H_α be the coroot for α , regarded as an element of $\text{Lie } \mathbf{T}$, then fix $X_{-\alpha}$ by requiring that $[X_\alpha, X_{-\alpha}] = H_\alpha$. Set $n(\alpha) = \exp(X_\alpha) \exp(-X_{-\alpha}) \exp(X_\alpha)$.
- (2) When $w \in \Omega$, $w \neq 1$, let $w = s_{\alpha_1} \dots s_{\alpha_r}$ be a reduced form, then set $n(w) = n(\alpha_1) \dots n(\alpha_r)$. It can be proved that $n(w)$ is independent of the choice of reduced expression for w . See Springer 9.3.2.
- (3) If $w = 1$ set $n(w) = 1$.

Thus if $\theta = w \rtimes \rho$, then define $n(\theta) = n(w) \rtimes \rho$ and

$$n(\theta_1)n(\theta_2) = t(\theta_1, \theta_2)n(\theta_1\theta_2), \quad \theta_i \in \Omega \rtimes \Gamma,$$

where $t(\theta_1, \theta_2)$ is a 2-cocycle of $\Omega \rtimes \Gamma$ in $\mathbf{T}(\bar{k})$

Lemma 2.1.

$$t(\theta_1, \theta_2) = \prod_{\substack{\alpha > 0 \\ \theta_1^{-1}\alpha < 0 \\ \theta_2^{-1}\theta_1^{-1}\alpha > 0}} (-1)^{\alpha^\vee}$$

Gauge. In this section, we will denote:

- X is a free finitely generated $\mathbb{Z}$ -module.
- Σ is a group, with action on X , and Σ contains an element ε such that $\varepsilon\lambda = -\lambda$, $\forall \lambda \in X$. (We will set $\Sigma = \{1, \varepsilon\} \times \Gamma$).
- Extend by trivial action on $k^\times$, Σ acts on $k^\times \otimes X$.
- $\mathcal{R}$ is a finite Σ -stable subset of X .
- p is a gauge on $\mathcal{R}$, that is $p : \mathcal{R} \rightarrow \{\pm 1\}$ and $p(-\lambda) = -p(\lambda)$, $\forall \lambda \in \mathcal{R}$.

We abbreviate the notation: Let $\sigma, \tau \in \Sigma$, then the product $\prod_{1, \sigma, \tau}^p$ is over $\lambda \in \mathcal{R}$

such that

$$p(\lambda) = 1, \quad p(\sigma^{-1}\lambda) = 1, \quad p(\tau^{-1}\sigma^{-1}\lambda) = 1.$$

Lemma 2.2.

$$t_p(\sigma, \tau) := \prod_{1, \sigma, \tau}^p (-1)^\lambda \in k^\times \otimes X.$$

is a 2-cocycle. This is just a form for now, but have explicit meaning in our cases.

Lemma 2.3. If q is also a gauge on $\mathcal{R}$, then t_q is cohomologous to t_p .

Definition 2.1. When $\Sigma = \{1, \varepsilon\} \times \Gamma$, and Σ acts transitively on $\mathcal{R}$. Then it has two Γ -orbits $\mathcal{R} = \mathcal{O} \cup -\mathcal{O}$ or Γ also acts transitively on $\mathcal{R} = \mathcal{O}$. In the former case $\mathcal{O}$ is called **asymmetric** and in the latter **symmetric**.

Lemma 2.4. Suppose that $\mathcal{R} = \mathcal{O} \cup -\mathcal{O}$ or Γ is the asymmetric case. Then the restriction of t_p to Γ is trivial.

Explicit Splitting. Suppose that p and q are gauges on $\mathcal{R}$ then t_p and t_q are cohomologous 2-cocycles of Σ in $k^\times \otimes X$. Then we can realize t_p/t_q as the differential of some 1-cocycle. See (2.4) of [LS] for the construction of $s_{p/q}$. We have

Lemma 2.5.

$$\partial s_{p/q} = t_p/t_q.$$

Lemma 2.6.

- (1) $s_{q/p}$ is cohomologous to $s_{p/q}$.
- (2) $s_{p/q}s_{q/r}$ is cohomologous to $s_{p/r}$.

2.2. a -data. Given the action of Σ on $k^\times$ with ε acting trivially. $\Gamma = \Gamma_k$. An a -datum for the action of Γ in $\mathcal{R}$ is a collection $\{a_\lambda : \lambda \in \mathcal{R}\}$, where

- (1) $a_\lambda \in \bar{k}^\times$, and $a_{\sigma\lambda} = \sigma a_\lambda$, $\forall \sigma \in \Gamma$, $\lambda \in \mathcal{R}$.
- (2) $a_{-\lambda} = -a_\lambda$.

We write $\prod_{\sigma, \tau}^p$ for the product over $\lambda \in \mathcal{R}$ such that $p(\sigma^{-1}\lambda) = 1$, $p(\tau^{-1}\sigma^{-1}\lambda) = 1$.

When a -data exist, choose one and set

$$u_p(\sigma) = \prod_{1, \sigma}^p a_\lambda^\lambda, \quad \sigma \in \Gamma.$$

Lemma 2.7.

$$t_p(\sigma, \tau) = \partial u_p(\sigma, \tau).$$

(This is independent of the choice of a -data.)

Now suppose that $\{b_\lambda : \lambda \in \mathcal{R}\}$ satisfies $b_\lambda \in \bar{k}^\times$ and

- (1) $b_{\sigma\lambda} = \sigma b_\lambda$, $\forall \sigma \in \Gamma$, $\lambda \in \mathcal{R}$,
- (2) $b_{-\lambda} = b_\lambda$.

There is no standard name for this and we could temporarily call this a b -datum, it measures the difference between different a -data.

Lemma 2.8. Let

$$v_p(\sigma) = \prod_{1, \sigma}^p b_\lambda^\lambda.$$

It is a 1-cocycle of Γ in $\bar{k}^\times \otimes X$.

Lemma 2.9. With the above notation, we have

- The class of v_p is independent of p .
- If $\mathcal{R} = \mathcal{O} \cup -\mathcal{O}$ is the asymmetric case, then v_p is cohomologically trivial.

Applications. Lets recall that we k a field of characteristic zero with algebraic closure $\bar{k}$, G is a connected reductive algebraic group defined and quasi-split over k , define:

- T is a maximal torus in G defined over k .
- $\{a_\alpha\}$ is a set of a -data for T . That is, for the action of $\Gamma_k = \text{Gal}(\bar{k}/k)$ on the set $R(G, T)$ of roots. In this case a -data exist.
- $(\mathbf{B}, \mathbf{T}, \{X_\alpha\})$ of G is a k -splitting. We can assume G is semisimple and simply-connected to ease notation.
- σ is the action of $\sigma \in \Gamma = \text{Gal}(\bar{k}/k)$ on $\mathbf{T}$, and $\Gamma = \{\sigma : \sigma \in \Gamma\}$. (So Γ preserves this k -splitting).
- Ω is the Weyl group $\Omega(G, \mathbf{T})$.
- Choose a Borel subgroup B of G (not necessarily defined over k) containing T , and $h \in G$ such that $(B, T)^h = (\mathbf{B}, \mathbf{T})$.
- Notate σ_T for the action of $\sigma \in \Gamma$ on T , it is transported to an action on $\mathbf{T}$ by $\text{Int}(h^{-1})$, we use the same notation σ_T for this action.
- Set $\Gamma_T = \{\sigma_T : \sigma \in \Gamma\} \subset \Omega \rtimes \Gamma$.
- Write $\omega_T(\sigma) \in \Omega$ defined by $\text{Int}(h^{-1}\sigma(h))$, then it can be shown that $\sigma_T = \omega_T(\sigma) \rtimes \sigma$.

The a -data for T also serves for the action of Γ on $R^\vee(G, T)$. After transport, it gives the action of Γ_T on $\mathcal{R} := R^\vee(G, \mathbf{T})$. By previous discussion,

$$n(\sigma_T) = n(\omega_T(\sigma)) \rtimes \sigma, \quad \sigma \in \Gamma.$$

and this satisfies

$$n(\sigma_T)\sigma(n(\tau_T))n((\sigma\tau)_T)^{-1} = \partial x(\sigma_T, \tau_T)^{-1}$$

where

$$x(\sigma_T) = \prod_{1, \sigma}^p a_\alpha^{\alpha^\vee}.$$

The gauge is defined such that $p(\alpha) = 1$ if and only if α is a root of $\mathbf{T}$ in $\mathbf{B}$. Then

$$\sigma_T \rightarrow x(\sigma_T) \cdot n(\omega_T(\sigma)) \rtimes \sigma$$

is a homomorphism of Γ_T in $N(\mathbf{T}(k), G(k)) \rtimes \Gamma$, and

$$\sigma_T \mapsto m(\sigma_T) := x(\sigma_T) n(\omega_T(\sigma))$$

is a 1-cocycle of Γ_T in $N(\mathbf{T}(\bar{k}), G(\bar{k}))$. Note that $hm(\sigma_T)\sigma(h^{-1})$ lies in $T(\bar{k})$, thus for given k -splitting, a -data, and $B \supset T$, it gives a well defined element $\lambda(T) = \lambda_{a_\alpha}(T)$ of $H^1(\Gamma, T(\bar{k})) = H^1(T)$. In the following we will analyze how those data affect this cohomology class.

• *Twist of the splitting.* The splitting $(\mathbf{B}, \mathbf{T}, \{\mathbf{X}_\alpha\})$ may be replaced only by $(\mathbf{B}^g, \mathbf{T}^g, \{\mathbf{X}_\alpha^g\})$, where $g \in G(\bar{k})$ such that $g\sigma(g)^{-1}$ lies in the center $Z(G)$ for all $\sigma \in \Gamma$. In this case, $\lambda(T)$ is multiplied by the class $\mathbf{g}$ in $H^1(T)$ represented by $\sigma \mapsto g\sigma(g)^{-1}$.

• *Changing a -data.* Suppose that the a -data $\{a_\alpha\}$ are replaced by $\{a'_\alpha\}$, where $a'_\alpha = b_\alpha a_\alpha$, then $\{b_\alpha\}$ is b -data we have discussed. In this case $\lambda_{a'_\alpha}(T) = \mathbf{b} \cdot \lambda_{a_\alpha}(T)$, here $\mathbf{b}$ is the class of b_q in $H^1(T)$, where $b_q : \sigma \mapsto \prod_{1, \sigma}^q b_\alpha^{\alpha^\vee}$, the class $\mathbf{b}$ is independent of gauge by previous lemma.

• *The choice of B .* It can be shown that $\lambda(T) \in H^1(T)$ is independent of the choice of $B \supset T$. The proof is not short and this is section (2.3.3) in [LS].

• *Twist of the torus T .* Suppose that $\text{Int}(g^{-1})$ maps T to T' over k and carries a -data $\{a_\alpha\}$ for T to $\{a'_\alpha\}$ for T' . Let $\mathbf{g} \in H^1(T)$ be the class of the cocycle $\sigma \mapsto g\sigma(g)^{-1}$ then $\lambda(T)$ is $\mathbf{g}$ times the image of $\lambda(T')$ under the homomorphism $H^1(T') \rightarrow H^1(T)$ given by $\text{Int}(g)$.

2.3. χ -data. In this section, we consider the case $k = \mathbb{C}$ and $\Gamma = \text{Gal}(L/F)$, where F is a local or global field and L is a finite Galois extension of F . $\Sigma = \langle \varepsilon, \Gamma \rangle$ acts on $\mathbb{C}^\times \otimes X$ with trivial action on $\mathbb{C}^\times$ the cocycle t_p is non-trivial in general. Let $W = W_{L/F}$ be the Weil group of L/F , and the action W on $\mathbb{C}^\times \otimes X$ through $W \rightarrow \Gamma$.

We define the χ -data for the action Γ on $\mathcal{R}$, we need to define some notations first, notate:

- $\Gamma_{+\lambda} = \{\sigma \in \Gamma : \sigma\lambda = \lambda\}$, $\Gamma_{\pm\lambda} = \{\sigma \in \Gamma : \sigma\lambda = \pm\lambda\}$, where $\lambda \in \mathcal{R}$. If λ is fixed, we delete it from the notations.
- $F_+ \subset L$ is the fixed field of Γ_+ , and $F_\pm \subset L$ is the fixed field of $\Gamma_\pm$. Note that $[F_+ : F_\pm]$ is 2 or 1 according to the Γ -orbit of λ is symmetric or asymmetric.
- $W_+ = W_{L/F_+}$ and $W_\pm = W_{L/F_\pm}$.

The χ -data for the action Γ on $\mathcal{R}$ is a collection $\{\chi_\lambda : \lambda \in \mathcal{R}\}$, where:

- (1) χ_λ is a character on $C_{+\lambda} = C_{F_{+\lambda}}$ the class group, which is $(F_{+\lambda})^\times$ when F is local and the idele-class group of $F_{+\lambda}$ when F is global.
- (2) Property one: $\chi_{-\lambda} = \chi_\lambda^{-1}$ and $\chi_{\sigma\lambda} = \chi_\lambda \circ \sigma^{-1}$, where $\sigma \in \Gamma$, $\lambda \in \mathcal{R}$.
- (3) Property two: If $[F_+ : F_\pm] = 2$ (symmetric case), then the character χ_λ on $C_{+\lambda}$ restricts to $C_{\pm\lambda}$ is the quadratic character corresponds to the quadratic extension $F_+/F_\pm$ from Class Field Theory.

Assume now Σ acts transitively on $\mathcal{R}$, the general case is the product of those what is defined in this case. Fix $\lambda \in \mathcal{R}$ now.

Choose representatives $\sigma_1, \dots, \sigma_n$ for $\Gamma_\pm \backslash \Gamma$, then $\pm\sigma_1^{-1}\lambda, \dots, \pm\sigma_n^{-1}\lambda$ are all elements of $\mathcal{R}$ without redundancy. Define a gauge p on $\mathcal{R}$ by

$$p(\lambda') = 1 \quad \text{if and only if} \quad \lambda' = \sigma_i^{-1}\lambda, \text{ for some } 1 \leq i \leq n.$$

Choose $w_1, \dots, w_n \in W$ such that w_i maps to σ_i under $W \rightarrow \Gamma$, then $w_1, \dots, w_n$ are representatives for $W_\pm \backslash W$. If $w \in W$ then define $u_i(w) \in W_\pm$ by

$$w_i w = u_i(w) w_{i'}, \quad 1 \leq i \leq n.$$

Choose $v_0 \in W_+$ and v_1 for $W_+ \backslash W_\pm$ in case $[F_+ : F_\pm] = 2$, or just choose $v_0 \in W_+$ when $F_+ = F_\pm$. For $u \in W_\pm$, define $v_0(u) \in W_+$ by

$$v_0 u = v_0(u) v_{i'}.$$

Set $s(u) = \chi_\lambda(v_0(u))$ for $u \in W_\pm$, and for $w \in W$, set

$$r_p(w) = \prod_{i=1}^n \chi_\lambda(v_0(u_i(w)))^{\lambda_i} = \prod_{i=1}^n s(u_i(w))^{\lambda_i},$$

where $\lambda_i = \sigma_i^{-1}\lambda$. Then r_p is a 1-chain of W with values in $\mathbb{C}^\times \otimes X$. If q is another gauge on $\mathcal{R}$, set

$$r_q = s_{q/p} r_p.$$

Lemma 2.10. *The coboundary of r_q is t_q , that is $\partial r_q = t_q$.*

The choices made no effect on r_q to up 1-coboundaries.

Now we define ζ -data, which is the analogue of b -data for a -data, for χ -data.

Lemma 2.11. *Suppose $\{\zeta_\lambda : \lambda \in \mathcal{R}\}$ satisfies:*

- (1) ζ_λ is a character on $C_{+\lambda}$ and hence on $W_{+\lambda}$. ($W^{\text{ab}} \cong C$)

- (2) $\zeta_{-\lambda} = \zeta_{\lambda}^{-1}$ for $\lambda \in \mathcal{R}$, and $\zeta_{\sigma\lambda} = \zeta_{\lambda} \circ \sigma^{-1}$ for $\sigma \in \Gamma$, $\lambda \in \mathcal{R}$.
- (3) If $[F_+ : F_{\pm}] = 2$ is the symmetric case, then ζ_{λ} is trivial on $C_{\pm\lambda}$.

Then

$$c(w) = \prod_{i=1}^n \zeta_{\lambda}(v_0(u_i(w)))^{\lambda_i}, \quad w \in W$$

is a 1-cocycle of W with values in $\mathbb{C}^{\times} \otimes X$. Its cohomology class is independent of the choices made in the construction.

Suppose that Σ is not transitive on $\mathcal{R}$, then we define r_q and c for each Σ -orbit, and then take products over all those orbits, the results are also denoted by r_q and c .

Applications. Let G be a connected reductive group defined and quasi-split over F . T is a maximal torus in G defined over F . Then we can choose $\{\chi_{\alpha}\}$ to be the χ -data for the action of $\text{Gal}(\bar{F}/F)$ on $R(G, T)$. There is no harm in replacing $\bar{F}$ by finite Galois extension $L \subset \bar{F}$ of F , over which T splits.

Fix a $\Gamma = \text{Gal}(\bar{F}/F)$ -splitting $(\hat{\mathbf{B}}, \hat{\mathbf{T}}, \{\mathbf{X}_{\alpha}^{\vee}\})$, we can realize ${}^L G = \hat{G} \rtimes W_F$, as before we can realize Γ as $\mathbf{\Gamma}$, which is the set of actions preserving the splitting.

Definition 2.2. A homomorphism $\xi : {}^L T \rightarrow {}^L G$ is an **admissible embedding** if:

- (1) ξ maps $\hat{T}$ isomorphically to $\hat{\mathbf{T}}$, and this isomorphism is the isomorphism attached to $(\hat{\mathbf{B}}, \hat{\mathbf{T}})$ and the choice of a Borel group B containing T .
- (2) $\xi(w) \in \hat{G} \times w$ for all $w \in W$.

It can be seen that the $\hat{G}$ -conjugacy class of ξ is independent of the choice $(\hat{\mathbf{B}}, \hat{\mathbf{T}})$ and B . We will give constructions of cochains from admissible embeddings.

Fix also B for now, to give an admissible embedding ξ , we have only to give a homomorphism $w \mapsto \xi(w) = \xi_0(w) \times w$, where $\xi_0(w) \in N(\hat{\mathbf{T}}, \hat{G})$. Let σ be the image of w via $W \rightarrow \Gamma$, then we can consider the action of σ , since the action of W defining L-groups factors through Γ . Then the action $\text{Int}(\xi(w))$ on $\hat{\mathbf{T}}$ should be induced from the action of σ on $\hat{T}$, through the isomorphism $\xi : \hat{T} \rightarrow \hat{\mathbf{T}}$. This action on $\hat{\mathbf{T}}$ is denoted by σ_T , as before we write it as $\omega_T(\sigma) \rtimes \sigma \in \Omega \rtimes \mathbf{\Gamma}$. Define

$$n(w) := n(\omega_T(\sigma)) \times w \in N(\hat{\mathbf{T}}, \hat{G}) \times w, \quad w \in W.$$

By the previous general lemma, we see it satisfies

$$n(w_1)n(w_2)n(w_1w_2)^{-1} = t_p(\sigma_1, \sigma_2),$$

where $w_i \mapsto \sigma_i$ under $W \rightarrow \Gamma$, the gauge $p(\alpha) = 1$ if and only if $\alpha^{\vee}$ is a root of $\hat{\mathbf{T}}$ in $\hat{\mathbf{B}}$. Thus with r_p the 1-cochain defined before, and

$$\xi(w) := r_p(w)n(\omega_T(\sigma)) \times w$$

We obtain an admissible embedding $\xi : {}^L T \rightarrow {}^L G$ by this. It is uniquely determined up to $\hat{\mathbf{T}}$ -conjugacy, the choice of borel subgroup B , and χ -data for T . We will examine how this admissible embedding changes when we change those data.

- *Twist of splitting.* Suppose that the splitting datum is replaced by $(\hat{\mathbf{B}}^g, \hat{\mathbf{T}}^g, \{\mathbf{X}_{\alpha}^{\vee g}\})$, where $g \in \hat{G}$, we may suppose g is Γ -invariant. Then the admissible embedding $\xi : {}^L T \rightarrow {}^L G$ is replaced by $\text{Int}(g^{-1}) \cdot \xi$, and the $\hat{G}$ -conjugacy class is not affected.

- *Change B .* It can be shown the $\hat{G}$ -conjugacy class of ξ is independent of the choice of B . See [LS] 2.6.2.

- *Change χ -data.* Suppose that the χ -data $\{\chi_\alpha\}$ is replaced by $\{\chi'_\alpha\}$, then we have $\chi'_\alpha = \zeta_\alpha \chi_\alpha$. Let c be the 1-cocycle defined earlier for ζ -data, then the admissible embedding ξ is replaced by $c \otimes \xi$, where

$$(c \otimes \xi)(t \times w) = c(w)\xi(c \times w).$$

- *Twist of T .* If $\text{Int}(g^{-1})$ maps T to T' over F , and carries $\{\chi_\alpha\}$ to $\{\chi'_\alpha\}$, then g defines a canonical isomorphism $\lambda_g : {}^L T' \rightarrow {}^L T$. And the admissible embeddings satisfy $\xi' = \xi \circ \lambda_g$.

3. TRANSFER FACTORS

3.1. Notations. In this section F will be a local field. And H, G, G^* are the original groups occurred in the Endoscopy data.

- $\gamma_H, \bar{\gamma}_H$ are strongly G -regular elements in $H(F)$ which are images (norms) of $\gamma_G, \bar{\gamma}_G$ strongly regular elements in G .
- $T_H, \bar{T}_H$ are torus containing $\gamma_H, \bar{\gamma}_H$ respectively. $T_G, \bar{T}_G$ are tori containing $\gamma_G, \bar{\gamma}_G$ respectively.
- Fix admissible embeddings $T_H \rightarrow T, \bar{T}_H \rightarrow \bar{T}$ of T_H and $\bar{T}_H$ in G^* . $\gamma, \bar{\gamma}$ be the images of $\gamma_H, \bar{\gamma}_H$ under those mappings.
- R is the root of T in G^* , $R^\vee$ is the coroot, Ω is the Weyl group. $R_H^\vee$ is the subsystem of $R^\vee$ of coroots from H via the admissible embedding, R_H is the roots from H , Ω_H is the Weyl group generated by R_H or $R_H^\vee$. The analogous notation for $\bar{T}$ is defined with bar added similarly.

We fix a -data, χ -data for the action of $\Gamma = \text{Gal}(\bar{F}/F)$ on the roots of T and $\bar{T}$. If the admissible embedding $T_H \rightarrow T$ is replaced by a $\mathfrak{U}(T)$ -conjugate, then we may transport the give a -data and χ -data to the new data.

We only have to consider the effects of the following:

- (1) Replacing $(T_H \rightarrow T, \{a_\alpha\}, \{\chi_\alpha\})$ by an $\mathfrak{U}(T)$ -conjugate.
- (2) Changing $\{a_\alpha\}$ and $\{\chi_\alpha\}$ with $T_H \rightarrow T$ fixed.

Let's analyze the dependence of definitions:

- $T_H \rightarrow T$ is given by the choice of B_H in H , B in G^* , pairs $(\mathcal{B}_H, \mathcal{T}_H)$ in $\hat{H}$, pairs $(\mathcal{B}, \mathcal{T})$ in $\hat{G}$, element $x \in \hat{G}$ such that $\text{Int}(x) \circ \xi$ that maps $\mathcal{T}_H$ to $\mathcal{T}$ and $\mathcal{B}_H$ into $\mathcal{B}$.
- $\hat{T}_H \rightarrow \mathcal{T}_H$ is given by the choice of (B_H, T_H) and $(\mathcal{B}_H, \mathcal{T}_H)$.
- $\hat{T} \rightarrow \mathcal{T}$ is given by the choice of (B, T) and $(\mathcal{B}, \mathcal{T})$.
- Endoscopy datum gives us $\xi : \hat{H} \rightarrow \hat{G}$, with $(\mathcal{B}_H, \mathcal{T}_H)$ and $(\mathcal{B}, \mathcal{T})$ fixed, we can choose one $x \in \hat{G}$.
- $\hat{T}_H \rightarrow \mathcal{T}_H \xrightarrow{\xi} \mathcal{T} \rightarrow \hat{T}$, defines the admissible embedding $T_H \rightarrow T$. (All arrows are isomorphisms, therefore invertible)

Now fix the splittings $(\mathcal{B}_H, \mathcal{T}_H, \{X^H\})$ of $\hat{H}$, and $(\mathcal{B}, \mathcal{T}, \{X\})$ of $\hat{G}$, and assume $(\mathcal{B}_H, \mathcal{T}_H)$ and $(\mathcal{B}, \mathcal{T})$ are the choices in $T_H \rightarrow T$ and $\bar{T}_H \rightarrow \bar{T}$. Up to equivalence of endoscopic data we may assume ξ maps $\mathcal{T}_H$ to $\mathcal{T}$ and $\mathcal{B}_H$ into $\mathcal{B}$.

The endoscopic datum s (the semisimple element) is central in $\xi(\hat{H})$, thus lies in $\mathcal{T}$ and its preimage in $\hat{T}_H$ is independent of the choice of B_H , so its image s_T in $\hat{T}$ depends only on the admissible embedding $T_H \rightarrow T$.

There is a canonical embedding of the center $Z(\hat{G})$ in $\hat{T}$. Let $\hat{T}_{\text{ad}} = \hat{T}/Z(\hat{G})$. Then $\pi_0 = \pi_0(\hat{T}_{\text{ad}}^\Gamma)$ denote the component group of Γ -invariants in $\hat{T}_{\text{ad}}$. The image of s_T in $\hat{T}_{\text{ad}}$ is Γ -invariant and thus defines an element s_T of π_0 .

Tate-Nakayama duality (one can see appendix in Renard) gives us a pairing:

$$\langle \cdot, \cdot \rangle : H^1(\Gamma, T_{sc}) \times \pi_0 \rightarrow \mathbb{C}^\times.$$

where T_{sc} denotes the induced torus of T in G_{sc}^* , the simply-connected covering of the derived group of G^* .

3.2. Δ_I .

Definition 3.1. $\Delta_I(\gamma_H, \gamma_G) := \langle \lambda(T), s_T \rangle$

Here the a -data and χ -data are transported to G_{sc}^* from G^* , and we have implicitly fixed some splitting **Spl** in G_{sc}^* to calculate $\lambda(T)$.

Lemma 3.1. $\Delta_I(\gamma_H, \gamma_G)/\Delta_I(\bar{\gamma}_H, \bar{\gamma}_G)$ is independent over the choice of **Spl**.

Now we check the effects. We have:

Lemma 3.2. If $(T_H \rightarrow T, \{a_\alpha\})$ is replaced by its g -conjugate, $g \in \mathfrak{U}(T_{sc})$ (we can choose T_{sc} here), then $\Delta_I(\gamma_H, \gamma_G)$ is multiplied by

$$\langle g_T, s_T \rangle^{-1}$$

where g_T is the class of $\sigma \mapsto g\sigma(g)^{-1}$ in $H^1(T_{sc})$.

Lemma 3.3. Suppose $\{a_\alpha\}$ is replaced by $\{a'_\alpha\}$ and $a'_\alpha = b_\alpha a_\alpha$, then $\Delta_I(\gamma_H, \gamma_G)$ is multiplied by

$$\prod_{\alpha} \chi_{\alpha}(b_{\alpha})$$

where the product is over representatives α for the symmetric orbits of Γ in the roots of T that are outside H .

3.3. Δ_{II} .

Definition 3.2.

$$\Delta_{II}(\gamma_H, \gamma_G) = \prod_{\alpha} \left(\frac{\alpha(\gamma) - 1}{a_{\alpha}} \right),$$

where the product is over representatives α of the orbits of Γ in the roots of T that are outside H .

Lemma 3.4. If $(T_H \rightarrow T, \{a_\alpha\}, \{\chi_\alpha\})$ is replaced by a $\mathfrak{U}(T)$ -conjugate then $\Delta_{II}(\gamma_H, \gamma_G)$ is unchanged.

Lemma 3.5. Suppose $\{a_\alpha\}$ is replaced by $\{a'_\alpha\}$ and $a'_\alpha = b_\alpha a_\alpha$, then $\Delta_I(\gamma_H, \gamma_G)$ is multiplied by

$$\prod_{\alpha} \chi_{\alpha}(b_{\alpha})^{-1}$$

where the product is over representatives α for the symmetric orbits of Γ in the roots of T that are outside H .

Suppose $\mathcal{O}$ is symmetric and q is a gauge on it. Define

- $X^{\mathcal{O}}$ free abelian group on set $\mathcal{O}_+$, with inherited action of Γ .
- $\mathcal{O}_+ = \{\alpha \in \mathcal{O} : q(\alpha) = 1\}$.
- X^{α} submodule generated by $\alpha \in \mathcal{O}_+$. So $X^{\mathcal{O}} = \text{Ind}_{\Gamma_{\pm}}^{\Gamma} X^{\alpha}$.
- $T^{\mathcal{O}}$ torus over F by $X^*(T_{\mathcal{O}}) = X^{\mathcal{O}}$, T^{α} torus over $F_{\pm\alpha}$ by $X^*(T_{\alpha}) = X^{\alpha}$, then T^{α} is one-dimensional anisotropic over $F_{\pm\alpha}$ and split over F_{α} (norm torus). It is seen that $T^{\mathcal{O}} = \text{Res}_F^{F_{\pm\alpha}} T^{\alpha}$.
- There is a natural homomorphism $T \rightarrow T^{\mathcal{O}}$ over F from the natural homomorphism $X^{\mathcal{O}} \rightarrow X^*(T)$, then $T(F) \rightarrow T^{\mathcal{O}}(F) \simeq T^{\alpha}(F_{\pm\alpha})$, let γ_{α} be the image of γ in $T^{\alpha}(F_{\pm\alpha})$.
- The norm map $T^{\alpha}(F_{\alpha}) \rightarrow T^{\alpha}(F_{\pm\alpha})$ is surjective and we may write $\gamma^{\alpha} = \delta^{\alpha} \bar{\delta}^{\alpha}$.

Lemma 3.6. *If the χ -data $\{\chi_\alpha\}$ are replaced by $\{\chi'_\alpha\}$ and $\chi'_\alpha = \chi_\alpha \zeta_\alpha$, then $\Delta_{II}(\gamma_H, \gamma_G)$ is multiplied by*

$$\prod^{asymm} \zeta_\alpha(\gamma^\alpha) \prod^{symm} \zeta_\alpha(\delta^\alpha)$$

where the products are both over representatives outside H , in asymmetric orbit pairs and symmetric orbits.

3.4. Δ_{III_1} or Δ_1 . When G is quasi-split, and $G^* = G$ with $\psi = \text{Id}$, since γ_H is norm of γ_G , then there is an $h \in G_{sc}$ such that $h\gamma_G h^{-1} = \gamma$. Set $v(\sigma) = h\sigma(h)^{-1}$, then the class $\text{inv}(\gamma_H, \gamma_G)$ of $v(\sigma)$ in $H^1(T_{sc})$ is well-defined.

Definition 3.3 (G quasi-split).

$$\Delta_1(\gamma_H, \gamma_G) = \langle \text{inv}(\gamma_H, \gamma_G), \mathbf{s}_T \rangle$$

In general, we work with the two pairs of elements γ_H, γ_G and $\bar{\gamma}_H, \bar{\gamma}_G$. There exist $h, \bar{h} \in G_{sc}^*$ such that

$$h\psi(\gamma_G)h^{-1} = \gamma \quad \text{and} \quad \bar{h}\psi(\bar{\gamma}_G)\bar{h}^{-1} = \bar{\gamma}.$$

Set

$$v(\sigma) = hu(\sigma)\sigma(h)^{-1} \quad \text{and} \quad \bar{v}(\sigma) = \bar{h}u(\sigma)\sigma(\bar{h})^{-1},$$

where $u(\sigma) \in G_{sc}^*$ and $\psi(\sigma)(\psi)^{-1} = \text{Int } u(\sigma)$, $\sigma \in \Gamma$. Then $v(\sigma)$ and $\bar{v}(\sigma)$ are cochains of Γ in T_{sc} and $\bar{T}_{sc}$, each well defined up to coboundaries because γ and $\bar{\gamma}$ are strongly regular.

Further, $\partial v = \partial \bar{v} = \partial u$, each coboundary taking values in the center Z_{sc} of G_{sc}^* . Let $U = U(T, \bar{T})$ be the torus

$$T_{sc} \times \bar{T}_{sc} / \{(z^{-1}, z) : z \in Z_{sc}\}.$$

Then $\sigma \rightarrow (v(\sigma)^{-1}, \bar{v}(\sigma))$ defines an element of $H^1(U)$ which is independent of the choices for $u(\sigma), h$ and $\bar{h}$. We write this class as

$$\text{inv} \left(\frac{\gamma_H, \gamma_G}{\bar{\gamma}_H, \bar{\gamma}_G} \right).$$

From $T_{sc} \times \bar{T}_{sc} \rightarrow U$ we get $H^1(T_{sc}) \times H^1(\bar{T}_{sc}) \rightarrow H^1(U)$. If G is quasi-split then $\text{inv} \left(\frac{\gamma_H, \gamma_G}{\bar{\gamma}_H, \bar{\gamma}_G} \right)$ is the image of $(\text{inv}(\gamma_H, \gamma_G)^{-1}, \text{inv}(\bar{\gamma}_H, \bar{\gamma}_G))$.

Recall that $\hat{T}_{\text{ad}}$ is $\hat{T}/Z(\hat{G})$, the torus dual to the preimage T_{sc} of T in G_{sc}^* . We denote by $\hat{T}_{sc}$ the torus dual to $T_{\text{ad}} = T/Z(G)$. Then the center $\hat{Z}_{sc}$ of the simply-connected covering of the derived group of $\hat{G}$, which is a finite group isomorphic to Z_{sc} , is canonically embedded in $\hat{T}_{sc}$ and $\hat{\hat{T}}_{sc}$. Set

$$\hat{U} = \hat{T}_{sc} \times \hat{\hat{T}}_{sc} / \{(z^{-1}, z) : z \in Z_{sc}\}.$$

Then $X^*(\hat{U}) \subset X^*(\hat{T}_{sc} \times \hat{\hat{T}}_{sc})$. At the same time, $X^*(U) \subset X^*(\bar{T}_{sc} \times \bar{\bar{T}}_{sc})$. The $\mathbb{Q}$ -pairing between $X^*(\hat{T}_{sc} \times \hat{\hat{T}}_{sc})$ and $X^*(\bar{T}_{sc} \times \bar{\bar{T}}_{sc})$ yields a dual $\mathbb{Z}$ -pairing between $X^*(\hat{U})$ and $X^*(U)$, and so $\hat{U}$ is the torus dual to U .

To the endoscopic datum s we attach $s_u \in \pi_0(\hat{U}^\Gamma)$ as follows. Suppose $\hat{s}$ lies in the preimage under $\mathcal{F}_{sc} \rightarrow \mathcal{F}_{\text{ad}}$ of the projection of s onto $\mathcal{F}_{\text{ad}} = \mathcal{F}/Z(\hat{G})$. From $\mathcal{F} \rightarrow \hat{T}$ and $\mathcal{F} \rightarrow \hat{\hat{T}}$ we obtain $\mathcal{F}_{sc} \rightarrow \hat{T}_{sc}$ and $\mathcal{F}_{sc} \rightarrow \hat{\hat{T}}_{sc}$. The images s_T and $s_{\bar{T}}$ of s depend only on the embeddings $T_H \rightarrow T$ and $\bar{T}_H \rightarrow \bar{T}$. The image $\mathbf{s}_U$ of $(s_T, s_{\bar{T}})$ in $\hat{U}$ is independent of the choice of s . It is also Γ -invariant and so defines an element $\mathbf{s}_U$ of $\pi_0(\hat{U}^\Gamma)$.

Definition 3.4.

$$\Delta_1(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G) = \left\langle \text{inv} \left(\frac{\gamma_H, \gamma_G}{\bar{\gamma}_H, \bar{\gamma}_G} \right), s_U \right\rangle.$$

Lemma 3.7. *If $T_H \rightarrow T$ and $\bar{T}_H \rightarrow \bar{T}$ are replaced by their g - and $\bar{g}$ -conjugates, where $g \in \mathfrak{U}(T_{sc})$ and $\bar{g} \in \mathfrak{U}(\bar{T}_{sc})$, then $\Delta_1(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G)$ is multiplied by*

$$\langle g_T, s_T \rangle \langle g_{\bar{T}}, s_{\bar{T}} \rangle^{-1},$$

where g_T is the class of $\sigma \rightarrow g\sigma(g^{-1})$ in $H^1(T_{sc})$, and $g_{\bar{T}}$ the class of $\sigma \rightarrow \bar{g}\sigma(\bar{g}^{-1})$ in $H^1(\bar{T}_{sc})$.

Note that if G is quasi-split over F then

$$\Delta_1(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G) = \langle \text{inv}(\gamma_H, \gamma_G), s_T \rangle^{-1} \langle \text{inv}(\bar{\gamma}_H, \bar{\gamma}_G), s_{\bar{T}} \rangle.$$

3.5. Δ_{III_2} or Δ_2 . We assume $\mathcal{H}$ is an L-group of H to make definition simpler, the general case uses z -pair.

For the construction here we fix Borel subgroups $B_H \supseteq T_H, B \supseteq T$ which yield the isomorphism $\hat{T}_H \rightarrow \mathcal{T}_H \rightarrow \mathcal{T} \rightarrow \hat{T}$ dual to $T_H \rightarrow T$. To the χ -data $\{\chi_\alpha\}$ are attached admissible embeddings $\xi_T : {}^L T \rightarrow {}^L G$ extending $\hat{T} \rightarrow \mathcal{T}$ and $\xi_{T_H} : {}^L T_H \rightarrow {}^L H$ extending $\hat{T}_H \rightarrow \mathcal{T}_H$. Then (identify ${}^L T$ with ${}^L T_H$ and use $\xi : \mathcal{H} \rightarrow {}^L G$)

$$\xi \circ \xi_{T_H} = a \xi_T,$$

where a is a 1-cocycle of W_F in $\mathcal{T}$ for the transport of the action of W_F on $\hat{T}$. We transport a to $\hat{T}$ without change in notation. Its class $\mathbf{a}$ in $H^1(W_F, \hat{T})$ is independent of the choice of B_H and B by (2.6.2) in [LS], and of the Γ -splittings $(\mathcal{B}_H, \mathcal{F}_H, \{X^H\})$ and $(\mathcal{B}, \mathcal{F}, \{X\})$ by (2.6.1) in [LS]. Further if $(T_H \rightarrow T, \{\chi_\alpha\})$ is replaced by its g -conjugate, $g \in \mathfrak{U}(T)$, then $\mathbf{a}$ is replaced by its image in $H^1(W_F, \hat{T}^g)$ under the map induced by $\text{Int } g^{-1}$ [see (2.6.4)].

Definition 3.5. $\Delta_2(\gamma_H, \gamma_G) = \langle \mathbf{a}, \gamma \rangle$.

The pairing is the usual one between $H^1(W, \hat{T})$ and $T(F)$ [Borel, Sect. 9].

Clearly, replacing $(T_H \rightarrow T, \{\chi_\alpha\})$ by an $\mathfrak{U}(T)$ -conjugate has no effect on $\Delta_2(\gamma_H, \gamma_G)$.

Lemma 3.8. *Suppose that the χ -data $\{\chi_\alpha\}$ are replaced by $\{\chi'_\alpha\}$, where $\chi'_\alpha = \chi_\alpha \zeta_\alpha$. Then $\Delta_2(\gamma_H, \gamma_G)$ is multiplied by*

$$\prod^{asymm} \zeta_a(\gamma^a)^{-1} \prod^{symm} \zeta_a(\delta^a)^{-1},$$

where the products are both over representatives outside H , in asymmetric orbit pairs and symmetric orbits.

3.6. Δ_{IV} and Δ . If $\gamma \in T(F)$ then $\prod_\alpha (\alpha(y) - 1)$, where the product is over all roots of T in G^* , lies in F . We set

$$D_{G^*}(y) = \left| \prod_\alpha (\alpha(y) - 1) \right|^{1/2}.$$

Definition 3.6. $\Delta_{IV}(\gamma_H, \gamma_G) = D_{G^*}(y) D_H(\gamma_H)^{-1}$.

Then $\Delta_{IV}(\gamma_H, \gamma_G)$ depends only on the stable conjugacy class of γ_H .

The Factor Δ . We now fix the pair $\bar{\gamma}_H, \bar{\gamma}_G$ and specify $\Delta(\bar{\gamma}_H, \bar{\gamma}_G)$ arbitrarily. Then we define

$$\Delta(\gamma_H, \gamma_G) = \Delta(\bar{\gamma}_H, \bar{\gamma}_G) \Delta(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G),$$

where

$$\Delta(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G)$$

is equal to

$$\frac{\Delta_I(\gamma_H, \gamma_G)}{\Delta_I(\bar{\gamma}_H, \bar{\gamma}_G)} \cdot \frac{\Delta_{II}(\gamma_H, \gamma_G)}{\Delta_{II}(\bar{\gamma}_H, \bar{\gamma}_G)} \cdot \frac{\Delta_{III_2}(\gamma_H, \gamma_G)}{\Delta_{III_2}(\bar{\gamma}_H, \bar{\gamma}_G)} \cdot \frac{\Delta_{IV}(\gamma_H, \gamma_G)}{\Delta_{IV}(\bar{\gamma}_H, \bar{\gamma}_G)} \cdot \Delta_{III_1}(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G).$$

In the case that G is quasi-split over F we set

$$\Delta_0(\gamma_H, \gamma_G) = \Delta_I(\gamma_H, \gamma_G) \Delta_{II}(\gamma_H, \gamma_G) \Delta_1(\gamma_H, \gamma_G) \Delta_2(\gamma_H, \gamma_G) \Delta_{IV}(\gamma_H, \gamma_G)$$

so that

$$\Delta(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G) = \frac{\Delta_0(\gamma_H, \gamma_G)}{\Delta_0(\bar{\gamma}_H, \bar{\gamma}_G)}.$$

Recall that $\Delta_I(\gamma_H, \gamma_G)$, but not $\Delta_I(\gamma_H, \gamma_G)/\Delta_I(\bar{\gamma}_H, \bar{\gamma}_G)$, depends on the choice of an F -splitting for G^* .

Theorem 3.1. $\Delta(\gamma_H, \gamma_G)$ is independent of the choice of admissible embeddings, a -data and χ -data.

Properties.

Lemma 3.9. $\Delta(\gamma_H, \gamma_G; \bar{\gamma}_H, \bar{\gamma}_G) = \Delta(\gamma_H, \gamma_G; \gamma_H, \gamma_G)$.

Lemma 3.10. $\Delta(\gamma_H, \gamma_G)$ depends only on the stable conjugacy class of γ_H in $H(F)$ and the conjugacy class of γ_G in $G(F)$.

4. FURTHER REMARKS

References.

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Remaining Problems.

- The concrete dependences of the definitions, on choices of constructing data.
- Specification of elements in G_{sc} and T_{sc} .
- Relation with Langlands correspondence?
- Motivation for different Δ ? Δ_{II} is for smooth extension, Δ_1 is to measure the weights for different stable orbits, Δ_2 correction to be compatible with Langlands correspondence. Δ_I Jump relations.